

Phase 7: Integration & External Access

Project Title: Student Success & Career Guidance Portal

7.1 Named Credentials (preferred for callouts)

When to use: Salesforce makes *callouts to external services* (LMS, Job APIs). **Named Credentials** centralize the endpoint + auth. Use them instead of hard-coding endpoints.

A. If external API is public or uses simple credentials

1. Setup → **Named Credentials** → **New Named Credential**.
2. Fill:
 - **Label:** LMS_NamedCred
 - **Name:** LMS_NamedCred
 - **URL:** https://api.mylms.com (replace with actual API root)
 - **Identity Type:** **Named Principal** (single identity for all callouts)
 - **Authentication Protocol:** **Password Authentication** (or **No Authentication** for public)
 - Enter username/password if using password auth.

3. Save.

The screenshot shows the 'LMS_NamedCred' configuration page in Salesforce Setup. The breadcrumb trail is 'SETUP > NAMED CREDENTIALS'. The page title is 'LMS_NamedCred'. There are 'Edit' and 'Delete' buttons in the top right corner. The form contains the following fields:

- Label:** LMS_NamedCred
- Name:** LMS_NamedCred
- URL:** https://github.com/settings/personal-access-tokens/8775376
- Enabled for Callouts:** A toggle switch that is currently turned on (indicated by a checkmark).
- Authentication:** A section with two options: 'External Credential' and 'No Authentication' (which is selected and highlighted in blue).
- Client Certificate:** A field for uploading a client certificate.
- Callout Options:** A section with four checkboxes:
 - Generate Authorization Header:** Checked.
 - Allow Formulas in HTTP Header:** Not checked.
 - Allow Formulas in HTTP Body:** Not checked.
 - Outbound Network Connection:** Not checked.
- Managed Package Access:** A section with two fields:
 - Created By Namespace:** A field for specifying the namespace.
 - Allowed Namespaces for Callouts:** A field for specifying allowed namespaces.

B. If external API uses OAuth 2.0 (recommended) — create an Auth. Provider first

1. Setup → **Auth. Providers** → **New**.

- **Provider Type:** choose the provider that matches the API (e.g., **GitHub** or the specific vendor if listed).
- **Name:** LMS_AuthProvider
- **Consumer Key / Secret:** get these from the external API developer console (not Salesforce).

- **Authorize Endpoint URL / Token Endpoint URL / User Info Endpoint URL:** paste the external API's OAuth URLs.
- **Default Scopes:** e.g., read write (depends on external API).
- Save.

The screenshot shows the 'Auth. Provider' detail page in Salesforce. The page has a 'SETUP' header and a 'Auth. Providers' section. Below this, there's a 'Auth. Provider Detail' section with a table of configuration details. To the right of the table are 'Edit', 'Delete', and 'Clone' buttons. Below the table is a 'Salesforce Configuration' section with another table of URLs. To the right of this table are also 'Edit', 'Delete', and 'Clone' buttons.

Auth. Provider Detail	
Auth. Provider ID	0SOgL000000rie1
Provider Type	GitHub
Name	LMS_AuthProvider
URL Suffix	LMS_AuthProvider
Consumer Key	https://github.com/settings/personal-access-tokens/8775376
Consumer Secret	Click to reveal
Authorize Endpoint URL	https://github.com/login/oauth/authorize
Token Endpoint URL	https://github.com/login/oauth/access_token
Default Scopes	read
Include Consumer Secret in SOAP API Responses	☒ ⓘ
Custom Error URL	
Custom Logout URL	
Portal	
Icon URL	
Use Salesforce MFA for this SSO Provider	☐ ⓘ

Salesforce Configuration	
Test-Only Initialization URL	https://orgfarm-081ad31f5a-dev-ed.develop.my.salesforce.com/services/auth/test/LMS_AuthProvider
Existing User Linking URL	https://orgfarm-081ad31f5a-dev-ed.develop.my.salesforce.com/services/auth/link/LMS_AuthProvider
OAuth-Only Initialization URL	https://orgfarm-081ad31f5a-dev-ed.develop.my.salesforce.com/services/auth/oauth/LMS_AuthProvider
Callback URL	https://orgfarm-081ad31f5a-dev-ed.develop.my.salesforce.com/services/auth/callback/LMS_AuthProvider
Single Logout URL	https://orgfarm-081ad31f5a-dev-ed.develop.my.salesforce.com/services/auth/rp/oidc/logout

2. After Save, on the Named Credential detail, click **Authenticate** (or **Authorize**). That opens the external provider's login page. Complete the OAuth flow and the Named Credential will store the access token/refresh token.

Use in Apex (callout):

```
HttpRequest req = new HttpRequest();
```

```
req.setEndpoint('callout:LMS_NamedCred/v1/courses'); // callout: + NamedCredential + path
```

```
req.setMethod('GET');
Http http = new Http();
HttpResponse res = http.send(req);
System.debug(res.getStatus());
System.debug(res.getBody());
```

Notes:

- Use callout:NamedCredentialName/path as the endpoint — Salesforce replaces host & auth.
- Named Credentials handle token refresh for OAuth providers automatically.

7.2 Platform Events — step-by-step

When to use: publish real-time notifications to external systems or internal subscribers (e.g., notify recruiter system when a student is shortlisted).

Create the Platform Event

1. Setup → **Platform Events** → **New Platform Event**.
 - **Label:** StudentShortlisted
 - **Object Name:** StudentShortlisted__e (platform events end with __e)
 - **Publish Behavior:** **Publish After Commit** (recommended)
 - **Save.**

The screenshot shows the Salesforce 'Platform Events' setup page. The header includes the 'SETUP' icon and the title 'Platform Events'. Below this, the specific event is named 'StudentShortlisted'. There are links for 'Standard Fields (4)' and 'Custom Fields & Relationships (4)'. The main section is titled 'Platform Event Definition Detail' and contains a table with the following information:

Platform Event Definition Detail		Description	
Singular Label	StudentShortlisted	Deployment Status	Deployed
Plural Label	StudentShortlisteds		
Object Name	StudentShortlisted		
API Name	StudentShortlisted__e		
Event Type	High Volume ⓘ		
Publish Behavior	Publish After Commit ⓘ		
Created By	Swayam Tamrakar, 9/23/2025, 2:49 AM		Modified By Swayam Tamrakar, 9/23/2025, 2:49 AM

At the top right of the page, there is a 'Help for this Page' link with a question mark icon. Above the table, there are 'Edit' and 'Delete' buttons.

2. Add fields: while on the event detail → **Fields & Relationships** → **New**:

- **Field 1:** StudentId__c — Data Type: Text (Length: 18)
- **Field 2:** JobId__c — Data Type: Text (Length: 18)
- (Add any other fields like ShortlistReason__c, ShortlistedBy__c)

Platform Event Definition Detail [Edit] [Delete]

Singular Label	StudentShortlisted	Description	
Plural Label	StudentShortlisteds	Deployment status	Deployed
Object Name	StudentShortlisted		
API Name	StudentShortlisted__e		
Event Type	High Volume [i]		
Publish Behavior	Publish After Commit [1]		
Created By	Swayam Tamrakar, 9/23/2025, 2:49 AM	Modified By	Swayam Tamrakar, 9/23/2025, 2:49 AM

Standard Fields

Action	Field Label	Field Name	Data Type	Controlling Field	Indexed
	<u>Created By</u>	CreatedBy	Lookup(User)		
	<u>Created Date</u>	CreatedDate	Date/Time		
	<u>Event UUID</u>	EventUuid	Text(36)		
	<u>Replay ID</u>	ReplayId	External Lookup		

Custom Fields & Relationships [New]

Action	Field Label	API Name	Data Type	Indexed	Controlling Field	Modified By
Edit Del	<u>JobId</u>	JobId__c	Text(18)			Swayam Tamrakar, 9/23/2025, 2:51 AM
Edit Del	<u>ShortlistedBy</u>	ShortlistedBy__c	Text(18)			Swayam Tamrakar, 9/23/2025, 2:51 AM
Edit Del	<u>ShortlistReason</u>	ShortlistReason__c	Text(18)			Swayam Tamrakar, 9/23/2025, 2:51 AM
Edit Del	<u>StudentId</u>	StudentId__c	Text(18)			Swayam Tamrakar, 9/23/2025, 2:50 AM

Triggers [New]

Action	Name	Api Version	Status	Size Without Comments	Last Modified By
Edit Del	<u>StudentShortlistedTrigger</u>	64.0	Active	493	Swayam Tamrakar, 9/23/2025, 2:55 AM

Subscriptions [i]

Action	Subscriber	Last Processed Id	Last Published Id	State	Batch Size	User
Manage	<u>StudentShortlistedTrigger</u>	19527	Not Available	Running	2,000	Automated Process

Publish a Platform Event (Apex example)

```
StudentShortlisted__e evt = new StudentShortlisted__e(  
    StudentId__c = '0033t00002ABCDE',  
    JobId__c = 'a0E3t00000XYZ12'  
);  
  
// Publish
```

```
Database.SaveResult sr = EventBus.publish(evt);  
System.debug('Published: ' + sr.isSuccess());
```

Subscribe to Platform Events

- **Apex trigger subscriber** (internal processing):

```
trigger StudentShortlistedTrigger on StudentShortlisted__e (after insert) {  
    for (StudentShortlisted__e evt : Trigger.New) {  
        // e.g., create a Task for Placement Officer  
        // evt.StudentId__c, evt.JobId__c  
    }  
}
```

Test: use Execute Anonymous to run the Apex publish code. Check subscriber behavior (e.g., a Trigger that creates a Task).

7.3 Remote Site Settings (when Named Credentials not used) — step-by-step

When to use: quick test callouts or older patterns — *less secure* than Named Credentials.

1. Setup → **Remote Site Settings** → **New Remote Site**.
 - **Remote Site Name:** mylms_api
 - **Remote Site URL:** https://api.mylms.com
 - **Description:** Allow callouts to LMS API

- Save.

Remote Site Settings

Remote Site Details

[Back to List: Custom Apps](#) [Help for this Page](#)

Remote Site Detail [Edit](#) [Delete](#) [Clone](#)

Remote Site Name	mylms_api	Modified By	Swayam Tamrakar, 9/23/2025, 2:59 AM
Remote Site URL	https://api.mylms.com		
Disable Protocol Security	☐		
Description	Allow callouts to LMS API		
Active	☒		
Created By	Swayam Tamrakar, 9/23/2025, 2:59 AM		

[Edit](#) [Delete](#) [Clone](#)

Use: In Apex callouts without Named Credential, you use the full https URL:

```
HttpRequest req = new HttpRequest();  
req.setEndpoint('https://api.mylms.com/v1/courses');  
req.setMethod('GET');  
Http http = new Http();  
HttpResponse res = http.send(req);
```

Recommendation: prefer **Named Credentials** — they centralize, secure, and make token refresh automatic.

Quick Testing checklist (do these after setup)

1. **Connected App:** verify Consumer Key & Secret are visible on the Connected App detail page. If you used Admin-approved, add your admin profile under **Manage Profiles**.
2. **Named Credential:** after creating + authenticating, run this sample from **Developer Console** → **Execute Anonymous**:

```
HttpRequest req = new HttpRequest();
```

```
req.setEndpoint('callout:LMS_NamedCred/v1/ping'); // replace with an endpoint that exists
```

```
req.setMethod('GET');
```

```
Http http = new Http();
```

```
HttpResponse res = http.send(req);
```

```
System.debug(res.getStatus() + ' ' + res.getBody());
```

3. **External Service:** build a simple Flow that calls one External Service action and run it.
4. **Platform Event:** execute the Apex publish snippet. Then check that your Subscriber Trigger executed (e.g., created a Task or wrote to debug log).
5. **Remote Site:** if using Remote Site, try a plain Apex callout to the external URL.